Architecture and Implementation Framework for AI-Assisted Document Verification and Regulatory Decision Support in Technical Education

Regulatory Framework and Operational Context

Regulatory governance of higher and technical education institutions requires managing large volumes of annual institutional filings covering Extension of Approval (EoA), variations in approved intake, the introduction of emerging supernumerary seats, institutional mergers, and establishment clearances for new technical campuses[1]. Under the statutory powers granted by the All India Council for Technical Education Act of 1987, the Council establishes procedural, legal, spatial, and financial standards across thousands of affiliated colleges and polytechnics throughout India[1]. Compliance mandates codified within the AICTE Approval Process Handbook (APH) necessitate evaluating diverse multi-page legal instruments, registered property deeds, architectural plans, fire and life safety clearances, fixed deposit receipts, and faculty employment rosters[1].

Historically, this administrative burden has been borne by decentralized regional offices and temporary regulatory bodies, including Scrutiny Committees, Re-Scrutiny Committees, Expert Visit Committees (EVC), Standing Hearing Committees (SHC), and the Standing Appellate Committee (SAC)[1]. Because these committees operate under strict statutory timelines before the start of each academic year, the manual review process encounters severe operational friction[1]. Scrutiny panels must evaluate hundreds of thousands of individual document pages, leaving the evaluation process vulnerable to oversights, forged certificates, unverified land-use conversions, and duplicated physical infrastructure[1].

Furthermore, decentralized manual checking creates regional variations in how strictly spatial and documentary norms are enforced[1]. At the same time, verifying documents across disparate state-level revenue authorities, municipal town planning bodies, affiliating universities, and corporate registries often proves impossible within standard committee review windows[1].

To resolve these administrative bottlenecks without compromising statutory oversight, this framework outlines an enterprise-grade, explainable artificial intelligence decision-support platform designed for direct integration with the AICTE web portal[1]. The proposed system does not replace statutory administrative bodies or exercise quasi-judicial authority. Instead, it serves as an automated triage, authentication, and compliance validation pipeline[1].

The platform normalizes incoming institutional filings, extracts structured key-value entities, verifies cryptographic signatures, detects localized forensic image manipulation, and reconciles spatial floor plans against APH infrastructure norms[6]. Operating strictly as an evidence-grounded decision-support system, the platform organizes applications into

risk-stratified verification channels, equipping evaluation committees with interactive visual dossiers and granular deficiency reports before formal hearings take place[1].

Core Functional Specifications and Compliance Capabilities

The system architecture is organized around six functional capabilities designed to satisfy the administrative standards of the AICTE Approval Process Handbook[1].

Intelligent Application Triage and Document Parsing

Upon submission through the institution-facing portal, the application engine identifies the filing category—such as an annual Extension of Approval based on self-disclosure, an expansion of academic intake, or the introduction of supernumerary seats for Non-Resident Indians (NRI) and Foreign Nationals (FN)—and loads the corresponding statutory checklist[1]. Heterogeneous uploaded files, which range from digitally generated vector PDFs to multi-generational scanned stamp papers, are ingested through a layout-aware optical character recognition and visual document understanding pipeline[5].

The extraction engine processes both unstructured text and spatial key-value relationships, identifying critical legal parameters such as Trust registration numbers, Khasra and Survey cadastral markers, FDR lien annotations, and Council of Architecture registration credentials[5]. Extracted parameters are instantly cross-referenced against historical AICTE databases to verify that declared physical land boundaries, program approvals, and paid Technical Education Regulatory (TER) charges match institutional records[5].

Forensic Document Analysis and Tamper Detection

Because static scans of municipal permits, land-use conversion certificates, and registered lease deeds are vulnerable to image manipulation, the platform incorporates multi-tiered forensic analysis[5]. For electronically signed PDF documents, the system inspects the document structure, validates embedded CMS and PKCS#7 signatures, traces X.509 certificate trust chains to authorized root authorities, and analyzes ByteRange allocations to flag any revisions made after the digital seal was applied[12].

When evaluating rasterized scans lacking cryptographic signatures, the platform deploys deep forensic neural networks to identify inconsistencies in image sensor noise, resampling patterns, copy-move duplications, and double JPEG compression artifacts[10].

To prevent cross-institutional asset fraud, facility photographs—including computer centers, language laboratories, advanced workshop equipment, and smart classrooms—are evaluated using perceptual hashing and vector embeddings, identifying visual assets that have been duplicated across distinct institutional files[6].

External Authority Verification Gateway

To eliminate reliance on easily manipulated paper records, the platform incorporates an asynchronous verification gateway that interfaces with external public data registries[11]. Extracted land parcel identifiers, sub-registrar deed references, and e-stamp certificates are validated through state land records systems (State *Bhulekh* and land registration departments)

to verify active ownership, lease tenures of thirty years or more, and uncontested educational land-use classification[6].

Corporate credentials for sponsoring Trusts, Societies, or Section 8 non-profit companies are validated against the Ministry of Corporate Affairs (MCA21) and the NITI Aayog NGO Darpan registry to verify legal standing and confirm authorized signatories[5].

Faculty credentials, principal appointments, and academic qualifications are verified through the Automated Permanent Academic Account Registry (APAAR), DigiLocker infrastructure, and state technical university portals, identifying unauthorized duplicate faculty listings across different institutions[18].

Spatial Infrastructure and Architectural Plan Analysis

Statutory compliance requires technical campuses to satisfy precise spatial standards across instructional, administrative, and amenities areas[6]. The platform automates spatial plan parsing by analyzing architectural blueprints certified by architects registered with the Council of Architecture[6]. Using computer vision models optimized for structural drawings, the system identifies the structural footprint of the building, isolating load-bearing walls, partition walls, and circulation boundaries[13].

By reading metric scale legends and dimension callouts, the spatial engine converts pixel boundaries into metric measurements to calculate the net carpet area for each distinct space[6]. These measurements are evaluated against APH norms, verifying whether classrooms meet the standard $66\ \mathrm{m}^2$ threshold for 60-student divisions, tutorial rooms provide the required $33\ \mathrm{m}^2$, drawing centers and computer CAD laboratories satisfy the $132\ \mathrm{m}^2$ minimum, and central libraries provide the mandated base reading area alongside proportional expansions for student populations exceeding 420[6].

In addition, interior photographs are analyzed with object-detection models to confirm the presence of required physical amenities, such as interactive smart boards, LCD projection equipment, and active safety systems[6].

Composite Risk Stratification and Workflow Routing

Rather than generating binary approval determinations, the system computes an aggregate anomaly score and sorts applications into three operational channels[1]. Applications that satisfy all spatial formulas, pass forensic checks, and match external registry records enter an accelerated administrative track[18]. Filings with clerical errors, minor spatial variances within permissible limits, or unconfirmed registry responses are routed to an intermediate clarification channel, which automatically compiles a structured deficiency memo for institutional rectification[1].

Applications exhibiting serious compliance red flags—such as invalidated cryptographic signatures, altered land-use documentation, duplicated campus photographs, or substantial infrastructure deficits—are routed to an escalation channel[1]. These files are prioritized for formal committee scrutiny and targeted physical inspection by the Expert Visit Committee[1].

Evidentiary Audit Trails and Explainable Scrutiny Dossiers

To maintain administrative defensibility, all algorithmic observations are compiled into an interactive evidence dossier[1]. The scrutiny interface displays extracted figures alongside source documents, marking flagged data with visual bounding boxes, displaying compression heatmaps over suspicious image areas, and generating side-by-side comparisons of institutional declarations against external registry records[10].

Every pipeline execution, rule check, model confidence metric, and officer review action is permanently recorded in an append-only, cryptographically linked audit log, establishing a complete chain of custody for regulatory reviews and legal appeals[1].

Detailed Technology Stack Specification and Trade-Off Analysis

Building an automated regulatory decision-support system requires balancing machine learning performance, data security, and system reliability. The table below details the selected enterprise components alongside the alternative options evaluated during architectural planning.

System Layer	Selected Technology	Alternative Considered	Primary Technical Justification
API & Service Layer	**FastAPI (Python 3.11+)**	Django REST Framework, Spring Boot, Node.js / Express	Native asynchronous event handling, unified Python ML execution, low serialization overhead via Pydantic, and automatic OpenAPI schema generation.
Document Processing & OCR	**PaddleOCR + LayoutLMv3**	Tesseract 5, Google Document AI, AWS Textract	Superior extraction accuracy across multi-column layouts and multilingual Indic scripts, support for self-hosted deployment to maintain sovereign data privacy, and unified spatial-textual

			multimodal embeddings.
Image & Forgery Forensics	**TruFor + pyHanko + ImageHash**	Vanilla Error Level Analysis (ELA), OpenCV Morphological Baselines	Pixel-level forgery localization combining RGB and learned noise patterns (Noiseprint++), supplemented by cryptographic CMS and PKCS#7 signature validation[10].
Spatial Infrastructure Analysis	**MitUNet (SegFormer + U-Net) + YOLOv8**	Vanilla Mask R-CNN, Classic Hough Transform	MitUNet captures global architectural contexts using transformer attention while preserving thin wall boundaries and structural openings via skip connections without geometric degradation[13].
Primary Relational Store	**PostgreSQL 16 (TimescaleDB / JSONB)**	Oracle Database, MySQL 8.0	Dynamic JSONB operations for evolving checklist schemas, ACID compliance, GIN indexing for high-speed queries, and native extension support that avoids complex multi-database architectures.
Vector & Similarity	**pgvector**	Pinecone, Milvus,	Eliminates

Store	**(Integrated within PostgreSQL)**	Qdrant	cross-network data synchronization between application storage and vector databases, scales to millions of document embeddings, and avoids cloud data egress dependencies.
Object & Document Storage	**MinIO (Self-Hosted S3-Compatible)**	AWS S3, Ceph, Local POSIX File System	High-throughput distributed object storage with complete S3 API parity, fine-grained bucket policies, full compliance with sovereign data residency requirements, and native object locking (WORM storage).
Task Queue & Orchestration	**Celery + Redis + Flower**	Apache Kafka, Apache Airflow, RabbitMQ	Lightweight task orchestration optimized for uneven ML computing workloads, robust retry mechanisms, lower infrastructure complexity than Kafka, and real-time task monitoring.
Frontend Interface	**React 18 + TypeScript + TailwindCSS**	Angular, Vue.js, Streamlit	Predictable state management, high performance when rendering

			high-resolution PDF canvases and SVG annotation overlays, and an extensive component ecosystem for complex administrative dashboards.

API and Microservice Engine

FastAPI on Python 3.11 provides the core API framework. While Django REST Framework (DRF) offers a mature ORM and built-in administrative tools, its synchronous nature introduces substantial I/O latency when handling large, multi-megabyte document submissions. FastAPI operates on an asynchronous event loop via Starlette and Uvicorn, allowing it to stream multipart payloads, write files to object storage, and dispatch message-queue tasks without exhausting the server thread pool.

Spring Boot provides enterprise-grade thread management and strict typing, but introduces architectural friction when interfacing with machine learning workloads. Managing neural network inference through Java Native Access (JNA) or separate gRPC microservices introduces serialization overhead and complicates deployment.

Similarly, Node.js handles asynchronous I/O well, but struggles with CPU-bound processing tasks, such as rasterizing PDF pages or computing cryptographic hashes. FastAPI unifies network I/O, asynchronous orchestration, and deep learning execution within a performant, maintainable Python environment.

Document Intelligence and Text Extraction

Document extraction utilizes a two-stage hybrid framework combining PaddleOCR with LayoutLMv3. Traditional OCR tools such as Tesseract 5 lack structural layout awareness, often producing disorganized text output when reading multi-column financial statements, land registration schedules, and architectural dimension tables.

Tesseract reads documents linearly, which leads to extraction errors when processing tables that lack clear cell borders. It also shows higher character error rates when reading low-resolution Indian legal documents that combine regional Indic scripts with Latin characters.





Proprietary cloud vision services, such as Google Document AI and AWS Textract, provide high extraction accuracy, but introduce recurring per-page API costs that grow expensive across thousands of multi-page filings. More importantly, routing institutional land deeds, structural certificates, and financial records to third-party public cloud endpoints conflicts with sovereign data residency standards for Indian government systems.

PaddleOCR delivers robust character recognition across complex layouts, including regional Indic scripts found on local land records (*7/12 extracts*, *Khata certificates*). The resulting bounding boxes and image patches are processed by LayoutLMv3, which models text, visual layout, and spatial position within a unified transformer architecture. This multimodal approach enables the system to reliably associate distant key-value pairs—such as matching a declared instructional area to a signature block elsewhere on the page—across varied institutional templates[6].

Forensic Anomaly and Forgery Engine

The forensic validation pipeline uses a multi-layered approach to evaluate document authenticity:





The pyHanko library processes electronically signed files, parsing the PDF structure to extract PKCS#7 digital signature blocks, validating certificate paths against authorized trust stores, and confirming that the document has not been altered after signing[12]. This catches manipulated approval records where visual details are modified while an invalid digital signature is retained[22].
Rasterized scans without digital signatures are analyzed using TruFor[10]. Traditional Error Level Analysis (ELA) flags differences in JPEG compression levels, but produces high false-positive rates on authentic scans that have undergone standard re-saving or downsampling.
TruFor addresses this by extracting a learned noise fingerprint (Noiseprint++) and analyzing it through a multi-scale transformer backbone, identifying localized inconsistencies caused by image splicing, digital retouching, or character substitution[10]. The model outputs both a localized anomaly heatmap and an overall tampering confidence score[10].
Submitted images are also processed using perceptual hashing (pHash) to generate 64-bit structural descriptors[16]. Comparing these hashes across the repository identifies cases where campus, laboratory, or workshop photographs have been duplicated across different institutional applications[6].

Spatial Infrastructure Analysis

Evaluating architectural blueprints requires deep structural parsing:





Traditional edge-detection techniques, such as Hough transforms, struggle on technical blueprints crowded with dimensions, hatching patterns, and text callouts[13]. Similarly, standard Mask R-CNN models often introduce discontinuities in thin, continuous structures like interior partition walls, resulting in open polygons that distort calculated areas[13].

The platform deploys MitUNet, a hybrid architecture combining a SegFormer transformer encoder with a lightweight CNN decoder and symmetrical skip connections[13]. Pre-trained on the CubiCasa5K dataset, the transformer layers capture broad structural context across the entire blueprint, while the convolutional decoder preserves thin walls, window openings, and door swings with high boundary accuracy[13].

A companion YOLOv8-nano model detects scale indicators and room text labels, establishing the pixel-to-meter conversion ratio and mapping identifiers to corresponding room boundaries[21]. The calculated carpet areas are then checked against the specific standards in the Approval Process Handbook[6].

Data Storage Architecture

Data persistence is split between an S3-compatible object store and an extended relational

database:



PostgreSQL 16 handles relational storage, institutional hierarchies, and application metadata. Its JSONB capabilities allow the system to adapt to changing approval checklists without requiring schema migrations[2]. Vector embeddings for text blocks and images are stored within the same database engine using the pgvector extension.

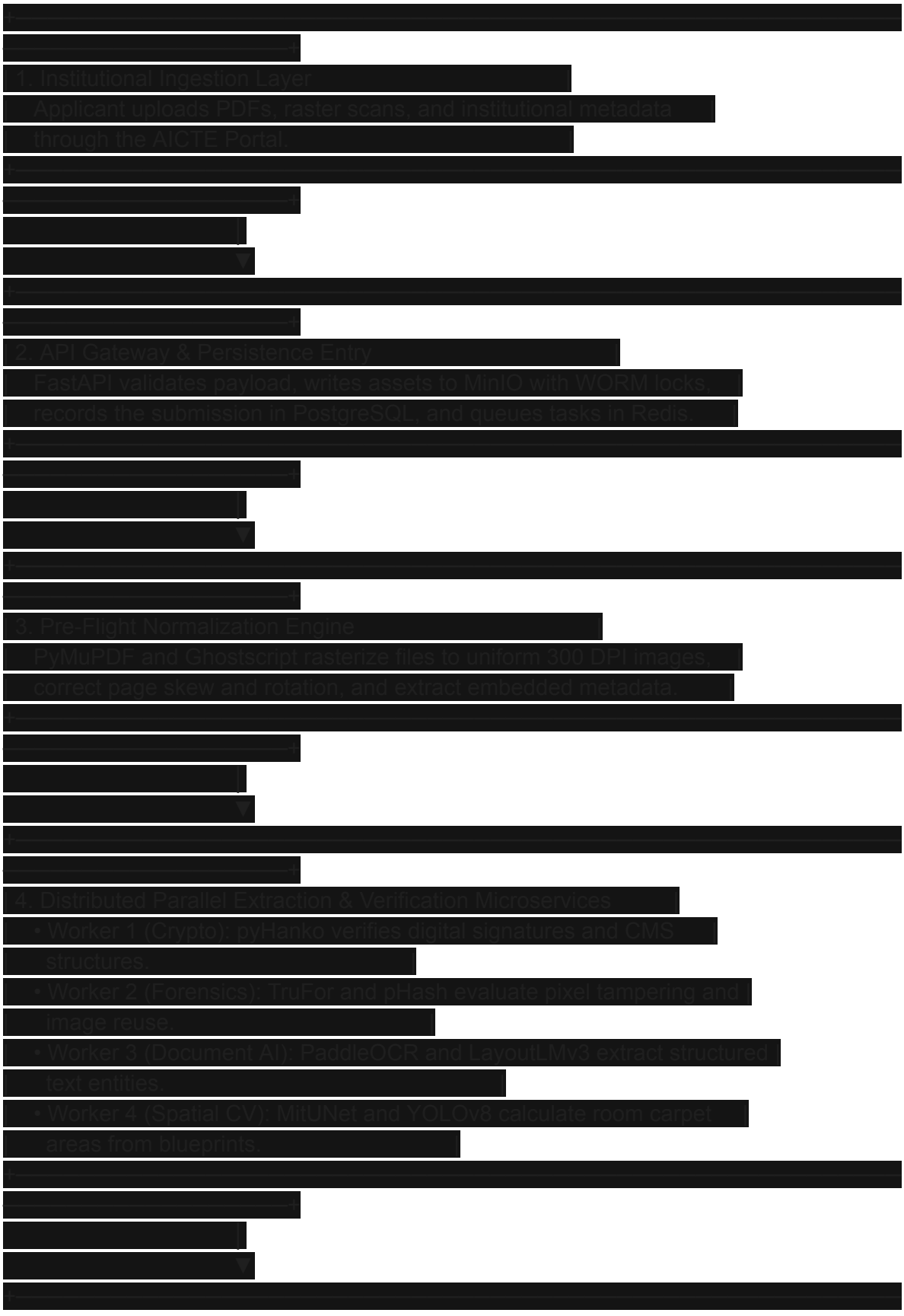
Using an integrated vector extension avoids the operational overhead and synchronization delays associated with running dedicated vector databases like Pinecone or Milvus. Storing vector embeddings alongside relational records simplifies disaster recovery, maintains transactional consistency, and ensures that institutional records and validation outputs are governed by the same ACID guarantees.

MinIO provides high-performance object storage for raw application uploads, generated page images, and diagnostic heatmaps. It runs on local server infrastructure, complying with data sovereignty requirements while avoiding public cloud egress costs.

Using native Write-Once-Read-Many (WORM) object locking, MinIO protects submitted documents from alteration or deletion during ongoing regulatory reviews, preserving the integrity of legal evidence[1].

Complete System Architecture and Operational Pipeline

The platform uses an asynchronous, microservices-based architecture to process document packages efficiently during high-volume filing cycles[1]. The complete data flow moves through eight integrated stages:


1. Institutional Ingestion Layer
Applicant uploads PDFs, raster scans, and institutional metadata through the AICTE Portal.
2. API Gateway & Persistence Entry
FastAPI validates payload, writes assets to MinIO with WORM locks, records the submission in PostgreSQL, and queues tasks in Redis.
3. Pre-Flight Normalization Engine
PyMuPDF and Ghostscript rasterize files to uniform 300 DPI images, correct page skew and rotation, and extract embedded metadata.
4. Distributed Parallel Extraction & Verification Microservices
• Worker 1 (Crypto): pyHanko verifies digital signatures and CMS structures.
• Worker 2 (Forensics): TruFor and pHash evaluate pixel tampering and image reuse.
• Worker 3 (Document AI): PaddleOCR and LayoutLMv3 extract structured text entities.
• Worker 4 (Spatial CV): MitUNet and YOLOv8 calculate room carpet areas from blueprints.



The system components interact across distinct operational stages to transform unstructured document uploads into structured compliance assessments:

1. **Submission and Ingestion:** The institution uploads required regulatory files via the AICTE portal, which submits the package to the FastAPI gateway[1]. The gateway generates a unique application tracking identifier, stores the raw files in MinIO storage with immutability locks enabled, creates an entry in PostgreSQL, and enqueues background processing tasks in Redis.

2. **Pre-Processing and Normalization:** Background Celery workers retrieve the files and execute pre-processing routines. Ghostscript and PyMuPDF convert multi-page PDFs into standardized 300 DPI PNG images, correct page orientation and skew, and extract document metadata.
3. **Parallel Analysis Pipelines:** Four specialized worker groups analyze the normalized files concurrently:
   - *Cryptographic Engine:* Inspects digitally signed documents using pyHanko, evaluating the validity of X.509 certificate chains and checking for post-signature document alterations[12].
   - *Forensic Engine:* Analyzes raster scans using TruFor to identify pixel-level splicing or image manipulation, and computes perceptual hashes (pHash) to flag duplicate visual assets[10].
   - *Document AI Engine:* PaddleOCR extracts text content and spatial bounding boxes across multilingual documents, while LayoutLMv3 classifies document sections and extracts key-value pairs into structured JSON.
   - *Spatial Analysis Engine:* MitUNet segments walls, openings, and rooms from architectural plans, while YOLOv8 identifies scale indicators and text labels, enabling the system to compute precise room carpet areas[6].
4. **External Registry Cross-Checking:** The system queries external authoritative APIs asynchronously using the extracted data, verifying property titles against state land registries, corporate status against MCA21, and staff credentials against APAAR and DigiLocker databases[11].
5. **Cross-Document Reconciliation:** An integrated business logic engine verifies consistency across submitted files, confirming that land areas listed on deeds match architectural certificates, and that planned classrooms support the requested student intake[5].
6. **Risk Triage and Routing:** The system computes an aggregate anomaly score ($S_{\text{composite}}$), assigning each application to an accelerated administrative path (Green), a deficiency clarification path (Amber), or a committee escalation path (Red)[1].
7. **Dossier Generation and Audit Logging:** The system compiles an interactive scrutiny dossier that displays extracted data alongside source documents with visual anomaly highlights, recording all processing steps and officer reviews in an immutable, hash-linked audit log[1].

The operational parameters for each processing stage are detailed in the pipeline overview table below.

Pipeline Stage	Input Artifacts	Core Algorithmic Operations	Execution Latency	Generated Output Artifacts
1. Ingestion &	Multipart PDF	File validation,	1.5 – 3.0 sec	Standardized

Invariant Pre-flight	streams, TIFF packages, Form metadata.	MIME typing, SHA-256 asset hashing, DPI normalization, PyMuPDF deskewing.	per 50 pages	300 DPI raster caches, normalized JSON metadata envelopes.
2. Cryptographic Validation	Native electronic PDFs, PKCS#7 detached signatures.	ByteRange verification, ASN.1 structure parsing, X.509 certificate validation via pyHanko[12].	150 – 400 ms per file	Certificate trust report, signature timestamp log, post-signing revision alert.
3. Forensic Tamper Screening	Scanned affidavits, structural certificates, photographs.	TruFor dense noise analysis, error level evaluation, pHash image comparison[10].	800 – 1,800 ms per image (GPU)	Pixel manipulation heatmaps, tamper probability scores, duplicate image links.
4. Document Parsing & OCR	Multi-page scanned forms, title deeds, FDR receipts.	PaddleOCR character recognition, LayoutLMv3 spatial entity and key-value extraction.	600 – 1,200 ms per page (GPU)	Normalized JSON entities (names, survey numbers, financial amounts, dates).
5. Spatial Plan Segmentation	Architectural blueprints (PDF/TIFF), CAD exports.	MitUNet wall and opening segmentation, YOLOv8 scale reading, polygon area calculation[13].	2.5 – 5.0 sec per floor plan	Vector room polygons, classified room labels, calculated carpet area tables.

6. External Gateway Queries	Khasra IDs, Corporate IDs, APAAR academic IDs[18].	Asynchronous REST and SOAP queries to State Bhulekh, MCA21, and DigiLocker endpoints[11].	1.0 – 6.0 sec (network dependent)	Authoritative registry validation records, mismatch alerts.
7. Cross-Record Reconciliation	Extracted entities, external registry data, portal records.	Declarative rule validation against APH norms, cross-document entity comparisons[6].	50 – 120 ms	Compliance verification matrices, structural deficiency summaries.
8. Triage & Dossier Generation	Multi-pipeline scores, anomaly heatmaps, parsed tables.	Evaluates composite anomaly formula, sorts applications into Green, Amber, or Red queues[1].	100 – 250 ms	Interactive scrutiny dossier, review queues, tamper-evident audit log entries.

Detailed Data Reconciliation and Validation Matrix

The platform's rule engine evaluates extracted parameters against specific compliance clauses defined across the Approval Process Handbook[1]. The table below outlines the primary reconciliation workflows executed across document categories.

Document Category	Primary Source Document	Extracted Data Fields	Cross-Reference Target	Regulatory Rule & Validation Criteria
Institutional Governance	Affidavit 1, Trust Deed, MoA[5]	Trust Name, Registration No., Trustee Names, Validity Date.	MCA21 Portal, State Charity Commissioner Registry, NGO Darpan.	Entity must be actively registered with educational operations

				authorized; trustee lists must match portal profiles[9].
Land Ownership & Title	Registered Sale / Gift Deed, Lease Deed[5]	Khasra/Survey Numbers, Plot Area, Land Use Category, Lease Duration.	State Land Records (Bhulekh), Town Planning Authorities.	Minimum contiguous land requirement met; educational land use certified; minimum 30-year lease validity for leased sites[6].
Building Infrastructure	Architect Certificate, Approved Plan[6]	Instructional Area, Administrative Area, Amenities Area, Total Plinth Area.	Architectural Floor Plans (MitUNet calculation)[13].	Instructional carpet area matches calculated polygon totals; classrooms $>$, tutorial rooms $>$ [6].
Safety & Occupancy	Structural Safety, Fire Safety Certificates[9]	Issuing Authority, License Validity, Building Height, Max Occupancy.	State Fire Service Portals, Municipal Issuing Records.	Document validity extends past the full academic operating year; safety approvals issued by certified municipal bodies[9].
Academic	University	Affiliated	AICTE	Intake

Operations	Affiliation, Previous EoA[1]	Courses, Approved Intake, Year of Expiry, Program Level.	Historical Database, Affiliating University Gazette[22].	numbers reconcile with portal declarations; program continuation permitted without regulatory debarment[1].
Financial Instruments	Fixed Deposit Receipts (FDR), Bank Guarantees[5]	FDR Number, Amount, Joint Account Holder, Maturity Date.	Bank API Gateways, Public Sector Core Banking Mechanisms.	Instrument held jointly with AICTE; principal deposit meets regulatory security threshold based on program tier[5].
Faculty & Staffing	Faculty Joining Reports, Salary Disclosures[29]	Aadhaar / APAAR ID, PAN, Designation, Scale of Pay, Qualifications[18] .	APAAR Platform, Employee Provident Fund (EPFO) APIs.	Faculty-to-Student ratio satisfies program norms (e.g., 1:15 or 1:20); prevents duplicate full-time faculty listings across colleges[29].

Mathematical Formulation of the Composite Anomaly Engine

To provide transparent and explainable recommendations, the platform combines individual validation results into a single composite anomaly score ($S_{\text{composite}}$) bounded between 0.0 and 1.0 :

$$S_{\text{composite}} = \min\left(1.0, \sum_{i=1}^{n} w_i \cdot \phi_i(D)\right)$$

In this formulation, w_i represents the regulatory weight assigned to a specific compliance check, while $\phi_i(D) \in [0, 1]$ represents the normalized anomaly score calculated for that factor within application dossier D. The sum of all weights equals one:

$$\sum_{i=1}^{n} w_i = 1.0$$

The composite score is calculated across four key verification dimensions:

Cryptographic and Document Integrity (ϕ_{crypto})

$$\phi_{\text{crypto}} = \frac{1}{|D_{\text{signed}}|} \sum_{d \in D_{\text{signed}}} \mathbb{I}\left(\text{Status}(d) \neq \text{VALID}\right)$$

Where $|D_{\text{signed}}|$ is the total number of electronically signed files, and $\mathbb{I}$ is an indicator function returning 1 if a signature check encounters an expired certificate, an untrusted root authority, or altered byte ranges[12].
Because an invalid digital signature represents a severe compliance issue, a single verified signature failure sets $\phi_{\text{crypto}} = 1.0$ with an assigned weight of $w_{\text{crypto}} = 0.35$.

Forensic Manipulation Probability (ϕ_{forensic})

$$\phi_{\text{forensic}} = \max_{img \in D} \left(P_{\text{tamper}}\left(img\right)\right)$$

This metric captures the highest tampering probability identified by the TruFor model across all submitted document pages and physical photographs[10]. The forensic check carries a weight of $w_{\text{forensic}} = 0.25$.

Spatial and Infrastructure Deficit (ϕ_{spatial})

$$\phi_{\text{spatial}} = \frac{1}{|R|} \sum_{r \in R} \max\left(0, \frac{A_{\text{required}}\left(r\right) - A_{\text{calculated}}\left(r\right)}{A_{\text{required}}\left(r\right)}\right)$$

Where R represents the required institutional spaces, such as classrooms, instructional laboratories, computer workshops, and libraries[6].

For each room r, the system calculates the percentage shortfall between the measured carpet area $A_{\text{calculated}}$ and the statutory requirement A_{required} [6]. The spatial analysis factor carries a weight of $w_{\text{spatial}} = 0.25$.

Cross-Record Data Contradiction ($\phi_{\text{reconcile}}$)

$$\phi_{\text{reconcile}} = \frac{\sum_{k \in K} \mathbb{I}\left(V_{\text{extracted}}(k) \neq V_{\text{verified}}(k)\right)}{|K|}$$

Where K represents the set of primary validation fields, including land parcel identifiers, Society registration numbers, total built-up areas, and faculty counts[5].

This factor measures the proportion of declared values ($V_{\text{extracted}}$) that conflict with verified external records (V_{verified})[11]. It carries an assigned weight of $w_{\text{reconcile}} = 0.15$.

Decision Triage Thresholds

- $S_{\text{composite}} < 0.15$ **(Accelerated Green Channel):** The application satisfies all core verification checks, exhibits verified digital signatures, and aligns with historical data. The file is recommended for expedited administrative sign-off[18].
- $0.15 \leq S_{\text{composite}} < 0.50$ **(Amber Clarification Channel):** The application exhibits minor discrepancies, such as slight spatial variances, unreadable scans, or pending external registry responses. The platform compiles an automated deficiency memo specifying the required rectifications[1].
- $S_{\text{composite}} \geq 0.50$ **(Red Escalation Channel):** The application exhibits significant compliance anomalies, such as invalid digital signatures, detected document tampering, duplicate campus photographs, or critical space deficits. The platform flags the application for detailed committee review and prioritizes the campus for on-site inspection by the Expert Visit Committee[1].

Regulatory Governance, Explainability, and Procedural Safeguards

To maintain compliance with Indian administrative law and statutory procedures, automated decision-support systems must avoid autonomous, unreviewable determinations[1]. Under Section 10 of the AICTE Act, 1987, the authority to approve, defer, or reject institutional applications rests with the Council and its statutory committees[1]. The platform incorporates specific procedural safeguards to ensure transparency and accountability.

When discrepancies are detected, the system generates an interactive scrutiny dossier rather than a simple warning flag[1]. If the forensic engine detects an altered building stability certificate, the evaluation interface presents the original document with bounding boxes outlining the affected area, displays the TruFor noise-distribution heatmap, and summarizes the issue in clear language (e.g., *"Localized noise variations and font mismatches detected within the structural engineer seal and signature block"*)[10].

This allows committee members to evaluate the underlying evidence directly rather than relying on an opaque automated score[1].

The platform avoids using generative Large Language Models for automated document extraction, preventing hallucination risks during compliance reviews. Text parsing relies on extractive token classification models that bind extracted data directly to character coordinates within the source file[6].

If scan degradation causes optical character recognition confidence to drop below 85% on critical legal documents, the engine bypasses automated processing and routes the file for manual human review[17].

Regulatory rules are maintained in an externalized configuration engine rather than embedded within machine learning models. When AICTE publishes revisions to the Approval Process Handbook, administrators can update validation rules and spatial thresholds directly without needing to retrain neural networks[6].

Data protection measures are implemented across all platform components. All processing workers, databases, and document stores operate on sovereign cloud infrastructure, such as the MeghRaj Government Cloud, ensuring that institutional records remain within Indian jurisdiction.

Personal identity details, including faculty Aadhaar numbers, are masked upon ingestion using cryptographic vaulting, allowing the system to verify staff counts while protecting personal privacy[17].

All validation runs, rule evaluations, and officer review decisions are recorded in an append-only log structured as an immutable cryptographic hash chain ($H_k = \text{SHA-256}(H_{k-1} \parallel \text{Event}_k)$), ensuring a complete, verifiable audit trail in the event of administrative appeals or legal proceedings[1].

Strategic Implementation Roadmap

Deploying the platform across regional offices involves four structured development phases to ensure system stability, calibrate regulatory rule sets, and support operational adoption[1].

Phase 1: Foundational Platform and Pipeline Ingestion

The initial phase establishes the core computing infrastructure and document extraction services. The containerized deployment environment is configured using FastAPI, MinIO object storage, Redis task queues, and PostgreSQL databases with pgvector enabled.

Engineers implement document normalization pipelines using Ghostscript and PyMuPDF to convert uploaded files into uniform 300 DPI image sets. The pyHanko library is integrated to validate X.509 certificates and PKCS#7 digital signature structures against authorized trust

stores[12].
PaddleOCR and LayoutLMv3 models are deployed to extract structured entities from core regulatory documents, including institutional affidavits, previous EoA records, Fixed Deposit Receipts, and fire safety certificates[1].

Phase 2: Forensic Analysis and External Registry Rails

The second phase implements forensic screening and external registry integrations. The TruFor deep learning pipeline is deployed on GPU workers to analyze scanned documents for pixel tampering, localized splicing, and compression anomalies[10].
Perceptual hashing (pHash) routines are integrated across the central database to flag duplicate campus and laboratory photographs submitted across different applications[6].
Asynchronous API connectors are developed to query external government databases, including state land records registries (*Bhulekh*), the Ministry of Corporate Affairs (MCA21) portal, and the DigiLocker and APAAR academic repositories[11].
A cross-document reconciliation engine is deployed to verify consistency between extracted values and historical AICTE records[5].

Phase 3: Spatial Computer Vision and Pilot Testing

The third phase deploys spatial blueprint analysis models and validates the system against historical application archives. MitUNet is trained on architectural floor plans, using weights pre-trained on the CubiCasa5K dataset and fine-tuned on technical campus blueprints[13].
A companion YOLOv8 model is implemented to detect scale legends and room labels, enabling automated calculations of net carpet areas for comparison against APH space requirements[6].
The platform is evaluated in a shadow validation pilot analyzing two thousand anonymized historical applications from previous scrutiny cycles[1]. Feedback from this pilot is used to calibrate anomaly score weights (w_i) and ensure automated triage recommendations align with past committee determinations[1].

Phase 4: Production Integration and Continuous Model Improvement

The final phase integrates the platform with the live AICTE portal across all regional offices[1].
The React-based scrutiny dashboard is deployed to regional officers and Scrutiny Committee members, enabling evidence-based application reviews and automated deficiency memo generation[1].
The three-tier triage routing engine is enabled to sort incoming filings into accelerated, clarification, and escalation review channels[1].
An active learning pipeline is established to collect anonymized edge-case documents and officer review feedback, supporting periodic model updates to ensure the system adapts to evolving document layouts and regional drawing standards over time[13].
The milestones, dependencies, and risk mitigation strategies for each phase are summarized in the roadmap table below.

Implementation	Primary	Technical	Risk Mitigation

Phase	Engineering Deliverables	Dependencies	Benchmarks
Phase 1: Foundations	FastAPI container clusters; MinIO storage with WORM locking; PDF normalization pipeline; PaddleOCR + LayoutLMv3 models.	Standardized GPU compute runtimes; validated CCA root trust stores[12].	Achieve $>$ extraction uptime and $<$ character error rate on standard legal stamp papers[5].
Phase 2: Forensics & APIs	TruFor forensic models; pHash image deduplication; asynchronous API connectors for DigiLocker, Bhulekh, and MCA21[10].	External API access agreements; secure outbound gateway configurations[17].	Keep false-positive manipulation flags below 4.0% on multi-generational institutional scans[10].
Phase 3: Spatial Plan AI	MitUNet blueprint segmentation models; YOLOv8 scale reading tools; pilot validation on 2,000 historical filings[13].	CubiCasa5K architectural dataset; labeled institutional blueprint archives[13].	Achieve $<$ area calculation variance compared to registered architect certificates[6].
Phase 4: Full Deployment	Live portal integration; React scrutiny dashboards; automated three-tier triage routing; active learning pipeline[1].	Regional officer training programs; approval from the competent regulatory authority[1].	Cut initial committee scrutiny times by 60% while maintaining a complete, defensible audit trail[1].

Synthesis and Regulatory Impact

The technical framework described in this report bridges the gap between regulatory

compliance requirements and applied artificial intelligence. By unifying layout-aware document extraction, cryptographic signature validation, neural forensic analysis, external registry cross-checking, and architectural segmentation within an asynchronous microservices architecture, the platform modernizes technical education governance[6].

Manual document checking is transformed from a time-consuming administrative bottleneck into an organized, evidence-driven supervisory workflow[1].

Most importantly, the platform preserves constitutional safeguards and administrative procedural fairness: automated microservices handle repetitive extraction and validation tasks, while statutory evaluation committees retain full authority over formal regulatory decisions[1].

Works cited

1. Approval Process Handbook 2021-22Revised - AICTE, https://www.aicte.gov.in/sites/default/files/Approval%20Process%20Handbook_2021-22%20(Revised).pdf
2. AICTE Approval Process Handbook 2024-25 to 2026-27 - S3waas, https://cdnbbsr.s3waas.gov.in/s35938b4d054136e5d59ada6ec9c295d7a/uploads/2025/03/2025031399.pdf
3. All India Council for Technical Education - Maharaja Surajmal Institute, https://www.msijanakpuri.com/wp-content/uploads/EOA-AICTE-upto-2025-2026.pdf
4. All India Council for Technical Education - Lloyd Business School, https://www.lloydbusinessschool.edu.in/images/LBS/website/pdf/aicte-approval-letter.pdf
5. India Council for Technical Education Approval Process Handbook, https://www.aicte.gov.in/downloads/Final-Approval-Process-Handbook-2017_18.pdf
6. AICTE Approval Process Handbook 2024-27 | PDF | Libraries - Scribd, https://www.scribd.com/document/806407077/AICTE-APH-Library
7. AICTE Approval Process Handbook 2024-27 | PDF - Scribd, https://www.scribd.com/document/685434943/AICTE-APH-2024-25-Draft
8. AICTE Approval Process 2025-2026 Report | PDF - Scribd, https://www.scribd.com/document/830491474/2-SOC1740039789-Scrutiny
9. APPROVAL PROCESS HANDBOOK - AICTE, https://www.aicte.gov.in/downloads/ApprovalProcessHandbook9Jan2010.pdf
10. TruFor: Leveraging All-Round Clues for Trustworthy Image Forgery, https://cvpr.thecvf.com/virtual/2023/poster/21548
11. Digital Identity Beyond Aadhaar: Blockchain-Based Identity in India, https://www.amulyacharan.com/2025/12/09/digital-identity-beyond-aadhaar-blockchain-based-identity-in-india/
12. How to Verify PDF E-Signatures Programmatically with Node.js, https://signb.ee/blog/verify-pdf-signature-programmatically
13. MitUNet: Enhancing Floor Plan Recognition using a Hybrid Mix, https://arxiv.org/html/2512.02413v1

14. pyhanko.sign.validation package - Read the Docs, https://pyhanko.readthedocs.io/en/latest/api-docs/pyhanko.sign.validation.html
15. image-forgery-detection · GitHub Topics, https://github.com/topics/image-forgery-detection
16. A script to find and delete duplicate images using pHash. - GitHub, https://github.com/philipbl/duplicate-images
17. Decentro Fabric & Flow APIs, https://postman.decentro.tech/
18. AICTE Approval Process Handbook 2024-27 | PDF - Scribd, https://www.scribd.com/document/691858591/APH-2024-27-Compressed
19. DigiLocker Issuer Setup & Integration Service - Meon Technologies, https://meon.co.in/digilocker-issuer
20. Marksheet Verification API | Instant Academic Record Validation, https://meon.co.in/marksheet-verification-api
21. GitHub - mageaustralia/FloorPlanAnalyzer: Experimental floor plan, https://github.com/mageaustralia/FloorPlanAnalyzer
22. All India Council for Technical Education - Satyam Fashion Institute, https://www.satyamfashion.ac.in/LOA-Report.PDF
23. TruFor: Leveraging All-Round Clues for Trustworthy Image Forgery, https://store.computer.org/csdl/proceedings-article/cvpr/2023/012900u0606/1PORjOasuUU
24. LLM-Integrated Semantic Deep Learning Framework for ... - MDPI, https://www.mdpi.com/2076-3417/16/13/6290
25. Saravanan Subramani an - CKCET, https://ckcet.edu.in/university-affln-and-aicte-eoa-1.pdf
26. TruFor - grip-unina - GitHub, https://github.com/grip-unina/TruFor
27. Papers/image-forgery/2023/trufor.md at main - GitHub, https://github.com/greatzh/Papers/blob/main/image-forgery/2023/trufor.md
28. Cascade Mask R-CNN and Keypoint Detection used in Floorplan, https://www.diva-portal.org/smash/get/diva2:1450823/fulltext01.pdf
29. Expert Evc - AAA College of Engineering, https://aaaenggcoll.ac.in/wp-content/uploads/2025/04/51-Approval-Document-for-Online-Scrutiny-Process.pdf
30. ADDENDUM/ CORRIGENDUM 2026-27 Of APH 2024-27 - AICTE, https://aicte.gov.in/downloads/aph/Final%20Addendum%202026-27.pdf
31. ADDENDUM/ CORRIGENDUM 2026-27 Of APH 2024-27, https://avantiscdnprodstorage.blob.core.windows.net/legalupdatedocs/52940/AICTE-issued-the-Corrigendum-regarding-the-Approval-Process-2026-27-February2026.pdf